VIT BHOPAL

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Project Report - Smart Airport Navigator

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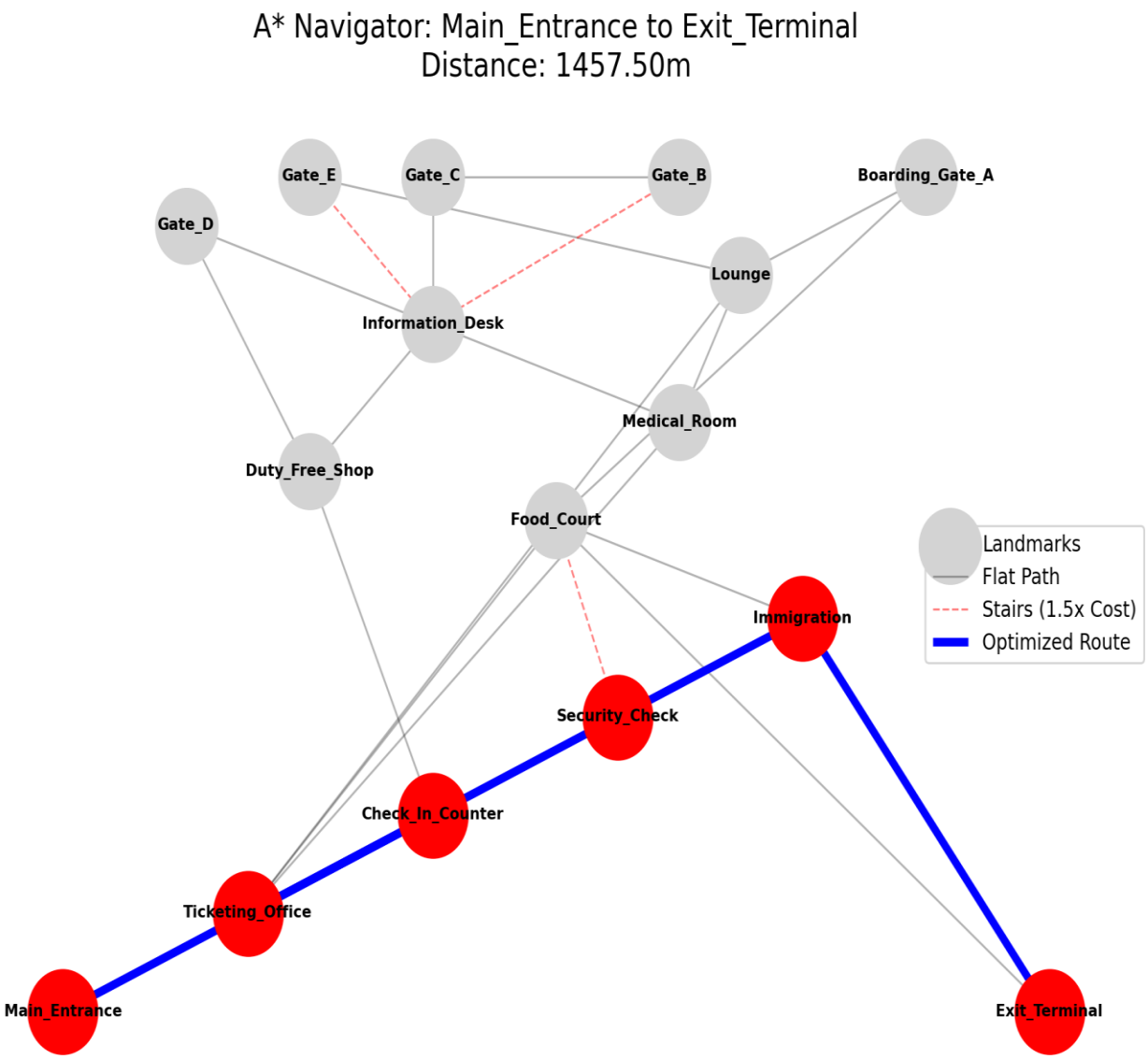
Domain - Artificial Intelligence

Tools – Python, Networkx, matplotlib, A*algorithm



1. ABSTRACT

Airports are large and complex environments where passengers often face difficulty in finding the shortest and most efficient routes between different locations such as check-in counters, boarding gates, lounges, and exits. This project presents a Smart Airport Navigator system that uses the A* (A-star) pathfinding algorithm to compute the optimal route between two selected points inside an airport.

The system models the airport as a graph using Python's NetworkX library, where nodes represent landmarks and edges represent paths with associated costs. The algorithm considers both distance and path type (flat or stairs), ensuring realistic navigation. A visual representation of the airport layout and the computed optimal path is generated using Matplotlib.

2. PROBLEM STATEMENT & MOTIVATION

Problem Statement-

Passengers in airports often:

- Get confused due to complex layouts
- Waste time finding boarding gates or services
- Face difficulty choosing the fastest path

There is a need for a system that:

- Provides shortest path navigation
- Considers real-world constraints like stairs

- Displays routes visually for easy understanding

Motivation-

The motivation behind this project is to:

- Improve passenger experience in airports
- Reduce navigation time and confusion
- Apply AI algorithms (A*) in real-world scenarios
- Demonstrate intelligent pathfinding using graphs

3. SYSTEM ARCHITECTURE & AI METHODOLOGY

System Architecture-

The system consists of the following components:

i)Input Module

- Takes starting point and destination from user

ii)Graph Model

Airport represented as a graph

- Nodes → Locations (e.g., Lounge, Gate A)
- Edges → Paths with weights

iii)A* Algorithm Module

Computes shortest path using:

- Actual cost (edge weights)
- Heuristic function (Euclidean distance)

iv)Visualization Module

- Displays airport layout
- Highlights optimal path

AI Methodology: A* Algorithm -

The A* algorithm is used because it is:

- Efficient
- Accurate
- Widely used in navigation systems

Working Principle

A* uses:

- g(n) = Actual cost from start to node
- h(n) = Heuristic estimate (Euclidean distance)
- f(n) = g(n) + h(n)

Heuristic Function

Euclidean Distance:

Special Feature in This Project

- Stairs have 1.5× cost, making them less preferred
- Ensures realistic path selection

4. IMPLEMENTATION DETAILS

Technologies Used

- Python
- NetworkX (Graph handling)
- Matplotlib (Visualization)
- Math library

The project is implemented in Python using the following libraries:
• NetworkX: Used for managing the graph data structure and calculating the A* path.

• Matplotlib: Used for the spatial visualization of the campus map.

5. RESULTS AND DISCUSSION

Results

The system successfully computes:

- Shortest path between any two points
- Total distance of the path

Visualization clearly shows:

- Airport layout
- Path types
- Optimal route

Example Outcome

If user selects:

- Start: Main_Entrance
- End: Boarding_Gate_A

The system:

- Calculates best route
- Avoids stairs if better flat path exists
- Displays highlighted route

Discussion

-Advantages

- Efficient pathfinding using A*
- Realistic navigation with weighted paths
- Easy-to-understand visualization
- Scalable to real airport systems

-Limitations

- Static layout (no real-time updates)
- No crowd or congestion handling
- Limited to predefined nodes

Future Enhancements

- Add real-time crowd data
- Mobile app integration
- Voice navigation support
- Wheelchair-friendly routing
- Indoor GPS integration

CONCLUSION

The Smart Airport Navigator demonstrates how AI algorithms like A* can be effectively used to solve real-world navigation problems. The system provides an optimized and user-friendly way to navigate complex airport layouts, improving efficiency and passenger experience.